SDGs 綠色亞運六大提案：解方、配套措施與具體作法（中英雙語）

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六組提案各有 1 個核心解方、4 項配套措施與具體作法；每項配套措施都用同一張評估矩陣打分，總分最高的優先執行。 Each of the six groups has one core solution, four supporting measures and concrete actions. Every measure is scored with the same matrix, and the highest scores go first.

評估矩陣怎麼看 How to Read the Matrix

每項配套措施依 5 個指標各打 1–5 分，滿分 25 分。5 分永遠代表「對提案最有利」。 Each measure gets 1–5 points on five criteria, 25 points in total. A score of 5 always means "best for the plan."

指標 Criterion	問什麼問題 Key question	5 分 =
可行性 Feasibility	現在的技術、法規、人力做得到嗎？ Can we do it with today's technology, rules and people?	馬上可以做 Ready to start now
成效 Effectiveness	能解決多少問題、減少多少碳排？ How much of the problem does it solve?	效果非常大 Very large impact
成本 Cost	要花多少錢？ How much money does it need?	花費很低 Very low cost
時程 Time	多久能完成？ How long does it take to set up?	6 個月內 Within 6 months
支持度 Support	政府、選手、居民、企業會接受嗎？ Will the government, athletes, residents and businesses accept it?	各方都支持 Everyone agrees

優先順序 Priority：20–25 分 = 優先執行 Do first；15–19 分 = 配合執行 Do with support；14 分以下 = 再評估 Rethink.

分數為課堂討論用的參考評估，各組可依自己的研究調整。 The scores are a starting point for class discussion. Each group can change them based on its own research.

第 1 組｜不興建大型一次性選手村，仍能提供完美的選手生活與醫療後勤

Group 1 | Great living and medical support for athletes without building a one-time athletes' village

核心解方 Core solution：分散式選手村。把既有飯店、大學宿舍和即將完工的社會住宅串成「選手村網絡」，再用既有醫院加上行動醫療站照顧選手。 A spread-out village. We link existing hotels, university dorms and new public housing into one "village network," and use local hospitals plus mobile clinics for medical care.

配套措施評估 Measures and Scores

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
A 征用飯店與大學宿舍 Use hotels and university dorms	5	4	4	5	4	22	優先 Do first
D 醫院合作＋行動醫療站 Partner hospitals + mobile clinics	5	4	4	4	5	22	優先 Do first
B 先蓋社會住宅，賽後轉出租 Build public housing first, rent it out later	4	5	3	2	5	19	配合 Do with support
C 可拆組模組屋 Movable modular homes	4	3	3	4	3	17	配合 Do with support

具體作法 Concrete Actions

1. 賽前 2 年與飯店、大學簽約，依場館位置分成 3–4 個「選手生活圈」，每圈距場館 20 分鐘內。 Sign contracts with hotels and universities two years early. Group them into 3–4 "athlete zones," each within 20 minutes of the venues.
2. 每個生活圈設一個服務中心：餐廳、訓練房、洗衣、物理治療集中在一處。 Set up one service center in each zone, with dining, a gym, laundry and physical therapy in one place.
3. 社會住宅按「選手村規格」先設計（無障礙、寬門、加長床），賽後 6 個月內交給市府出租。 Design public housing for athletes first (barrier-free, wide doors, long beds). Give it to the city to rent out within six months after the Games.
4. 與在地醫學中心簽協議，每個生活圈配 1 個行動醫療站，重症全部轉送綁定醫院。 Sign agreements with local medical centers. Put one mobile clinic in each zone and send serious cases to partner hospitals.
5. 用遠距醫療平台做第一線問診與翻譯，減少醫護人力壓力。 Use an online medical platform for first checks and translation, so doctors and nurses are not overloaded.

關鍵風險 Key risk：飯店旺季堵房。解法是提早簽約並寫入房價上限。 Hotels may be full in peak season. Sign early and set a price limit in the contract.

第 2 組｜為數萬名選手與觀眾提供低碳優質且安全的餐飲服務

Group 2 | Low-carbon, high-quality and safe food for tens of thousands of athletes and fans

核心解方 Core solution：在地、當季、植物優先的「綠色菜單」。食材以 200 公里內的台灣農產為主，每餐提供植物性蛋白選項，並用食安追溯系統把關。 A "green menu" that is local, seasonal and plant-forward. Most food comes from Taiwanese farms within 200 km, every meal has a plant-protein choice, and a tracking system keeps food safe.

配套措施評估 Measures and Scores

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
A 在地當季食材採購 Buy local, seasonal food	5	4	4	4	5	22	優先 Do first
B 可重複使用餐具＋押金制 Reusable tableware with a deposit	4	4	4	4	4	20	優先 Do first
C 食安追溯 QR Code Food-safety QR code tracking	4	3	4	4	5	20	優先 Do first
D 植物性蛋白為預設選項 Plant protein as the default choice	4	5	4	5	2	20	優先 Do first
E 剩食 AI 預測＋食物銀行 AI food-waste forecast + food bank	3	4	3	3	4	17	配合 Do with support

具體作法 Concrete Actions

1. 賽前 1 年與農會、產銷班簽契作合約，依比賽月份規劃當季菜單。 Sign farming contracts with local farmers' groups one year early, and plan the menu around the season of the Games.
2. 菜單標示每道菜的碳足跡（綠／黃／紅燈），植物性餐點放在取餐線最前面。 Show the carbon footprint of each dish with green, yellow and red labels, and put plant-based dishes first in the food line.
3. 選手餐廳保留運動營養師設計的高蛋白菜單，素食與肉食都符合體能需求。 Athlete dining halls keep high-protein menus designed by sports nutritionists, with both plant and meat options.

4. 每批食材的產地、檢驗報告、運送溫度上傳平台，掃餐盒 QR Code 即可查詢。 Upload the farm, test results and transport temperature of every batch. Scan the QR code on the box to check them.
5. 以購票人數預測備餐量，每日未售出的安全食物當晚送往食物銀行。 Use ticket sales to predict how much food to make. Send safe, unsold food to food banks the same night.

關鍵風險 Key risk：支持度最低的是植物性預設（D），部分選手與觀眾可能反彈。解法是「預設」不等於「強制」，肉食仍可自由選。 Plant protein as the default (D) has the lowest support, since some people may not like it. "Default" does not mean "forced": meat is still a free choice.

第 3 組｜場館與住宿分散時，實現零碳排、不延誤的精準接駁調度

Group 3 | Zero-emission, on-time shuttles when venues and hotels are far apart

核心解方 Core solution：「大眾運輸為主、電動接駁為輔、AI 調度中心統一指揮」。觀眾搭捷運、台鐵、高鐵，選手與工作人員搭電動巴士，所有車輛由同一個調度中心即時追蹤。 Public transport first, electric shuttles second, one AI control center for all. Fans take the MRT and trains; athletes and staff ride electric buses, and one center tracks every vehicle in real time.

配套措施評估 Measures and Scores

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
C 門票含當日大眾運輸免費 Free public transport with a ticket	5	5	3	5	5	23	優先 Do first
E 公共自行車與步行最後一哩 Shared bikes and walking for the last mile	5	2	5	5	4	21	優先 Do first
A 電動巴士專用接駁 Electric buses for athletes and staff	4	5	2	3	5	19	配合 Do with support
B 賽事專用道與號誌優先 Games lanes and priority traffic lights	4	4	4	4	3	19	配合 Do with support
D AI 即時調度中心 AI real-time control center	4	4	3	3	4	18	配合 Do with support

具體作法 Concrete Actions

1. 電子門票綁定當日捷運、公車、台鐵免費搭乘，場館不設一般停車場。 Link each e-ticket to free MRT, bus and train rides that day. Venues offer no regular parking.

2. 向客運業者租用電動巴士，賽後歸還繼續跑公車路線，不自己買車。 Lease electric buses from bus companies. After the Games, they go back to normal city routes, so no new buses are bought.
3. 選手生活圈到場館之間劃設賽事專用道，路口號誌對接駁車自動變綠燈。 Mark Games lanes between athlete zones and venues. Traffic lights turn green for shuttles automatically.
4. 調度中心用 GPS 與賽程資料預測人潮，比賽延長時自動加派車。 The control center uses GPS and the match schedule to predict crowds, and adds buses when a game runs late.
5. 每個場館留 2 輛備援車與急速充電樁，確保選手不會因為沒電而遲到。 Keep two backup buses and fast chargers at every venue, so athletes are never late because of a low battery.

關鍵風險 Key risk：電動巴士充電時間長、成本高（A 成本僅 2 分）。用「租而不買」與夜間集中充電降低風險。 Electric buses take time to charge and cost a lot (A scores only 2 on cost). Leasing instead of buying and charging at night lower the risk.

（提案原文「零探排」應為「零碳排」，此處已更正。）

第 4 組｜在不大量新建場館的前提下，使比賽營運過程達到淨零碳排

Group 4 | Net-zero carbon operations without building many new venues

核心解方 Core solution：「先少蓋、再用綠電、再節能，最後才抵換」。依這個順序減碳，碳排放全程盤查並公開，剩下減不掉的部分才用碳權抵換。 Build less, then use green power, then save energy, and offset only at the end. Every step is measured and shared with the public, and only the carbon we cannot cut is offset.

配套措施評估 Measures and Scores

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
A 優先用既有與臨時場館 Use existing and temporary venues first	5	5	5	4	4	23	優先 Do first
B 綠電憑證＋場館屋頂太陽能 Green power certificates + rooftop solar	4	5	3	3	5	20	優先 Do first
C 智慧能源管理（照明、空調） Smart energy control for lights and AC	4	4	3	4	4	19	配合 Do with support
D 碳盤查＋公開碳排儀表板 Carbon audit + public carbon dashboard	4	3	4	4	4	19	配合 Do with support
E 碳權抵換（最後手段） Carbon offsets (last step only)	5	2	3	5	3	18	配合 Do with support

具體作法 Concrete Actions

1. 盤點全台既有體育館、大學場地，先分配比賽項目；不夠的才用可拆解的臨時場館。 List all existing gyms and university venues and assign sports to them first. Use temporary venues that can be taken apart only when needed.
2. 既有場館的改建預算優先用在屋頂太陽能、LED 與高效率空調。 Spend the upgrade budget on rooftop solar, LED lights and efficient air conditioning first.
3. 賽事期間 100% 使用綠電（太陽能、離岸風電），以再生能源憑證（T-REC）證明。 Use 100% green power from solar and offshore wind during the Games, proven with renewable energy certificates (T-REC).
4. 依 ISO 14064 做碳盤查，官網設「即時碳排儀表板」，每日更新。 Measure carbon with the ISO 14064 standard and show a daily-updated carbon dashboard on the official website.
5. 減不掉的碳排放，優先購買台灣在地的自願減量額度（如造林、藍碳）。 For carbon we cannot cut, buy local Taiwanese carbon credits first, such as tree planting and blue carbon projects.

關鍵風險 Key risk：過度依賴碳抵換會被批評「漂綠」（E 成效僅 2 分）。所以抵換只能是最後一步，且要公開抵換清單。 Relying too much on offsets can look like greenwashing (E scores only 2 on effectiveness). Offsets must be the last step, with a public list.

第 5 組｜透過數位科技減少紙張浪費，並帶動百萬觀賽群眾一起參與減碳

Group 5 | Use digital tools to cut paper waste and get a million fans to help reduce carbon

核心解方 Core solution：一個官方 App 完成「買票、入場、導覽、減碳任務」。紙本票券與節目冊全部數位化，觀眾完成減碳行動可獲得積分換獎勵。 One official app for tickets, entry, guides and green missions. Paper tickets and programs go digital, and fans earn points for green actions.

配套措施評估 Measures and Scores

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
A 電子票券＋手機入場 E-tickets + phone entry	5	4	4	5	5	23	優先 Do first
E 數位包容：保留少量紙本服務台 Digital inclusion: a few paper help desks	5	2	4	5	5	21	優先 Do first
B 電子節目冊與 AR 場館導覽 E-programs and AR venue maps	5	3	4	4	4	20	優先 Do first

配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
C 減碳任務積分換獎勵 Green mission points for rewards	4	4	3	4	4	19	配合 Do with support
D 個人觀賽碳足跡計算 Personal carbon footprint calculator	4	3	4	4	3	18	配合 Do with support

具體作法 Concrete Actions

1. 所有票券只發電子票，入場掃 QR Code；節目冊、賽程表、地圖都放在 App。 Sell e-tickets only and scan a QR code at the gate. Put programs, schedules and maps in the app.
2. App 的減碳任務：搭大眾運輸、自備杯具、餐具回收，每項得分，可換紀念品或下一場優惠。 In-app green missions: take public transport, bring your own cup, return tableware. Each one earns points for souvenirs or discounts.
3. 全場觀眾的減碳總量即時顯示在場館大螢幕，形成「一起達標」的氣氛。 Show the total carbon saved by all fans on the stadium screen, so everyone feels they reach the goal together.
4. 各場館入口設 1 個服務台，為長者與沒有智慧型手機的人印製票券。 Put one help desk at each entrance to print tickets for older people and those without smartphones.
5. 賽後公布「省下多少張紙、幾公噸碳」的成果報告。 After the Games, publish a report on how much paper and carbon were saved.

關鍵風險 Key risk：全面數位化可能排除長者與弱勢族群。E 雖然減碳成效低，卻是公平參與的必要配套。 Going fully digital may leave out older or disadvantaged people. Measure E saves little carbon, but it is needed for fair access.

第 6 組｜賽事結束後，所有資材與設備 100% 循環利用，留下永續資產

Group 6 | Reuse 100% of materials and equipment after the Games and leave a lasting legacy

核心解方 Core solution：「買之前就想好去處」的循環設計。能租就不買；必須買的每一件都登錄「資產身分證」，賽前就指定賽後要捐給誰、移到哪裡。 Plan the second life of every item before buying it. Rent whenever possible; every item we must buy gets an "asset ID," and its next owner is chosen before the Games.

配套措施評估 Measures and Scores

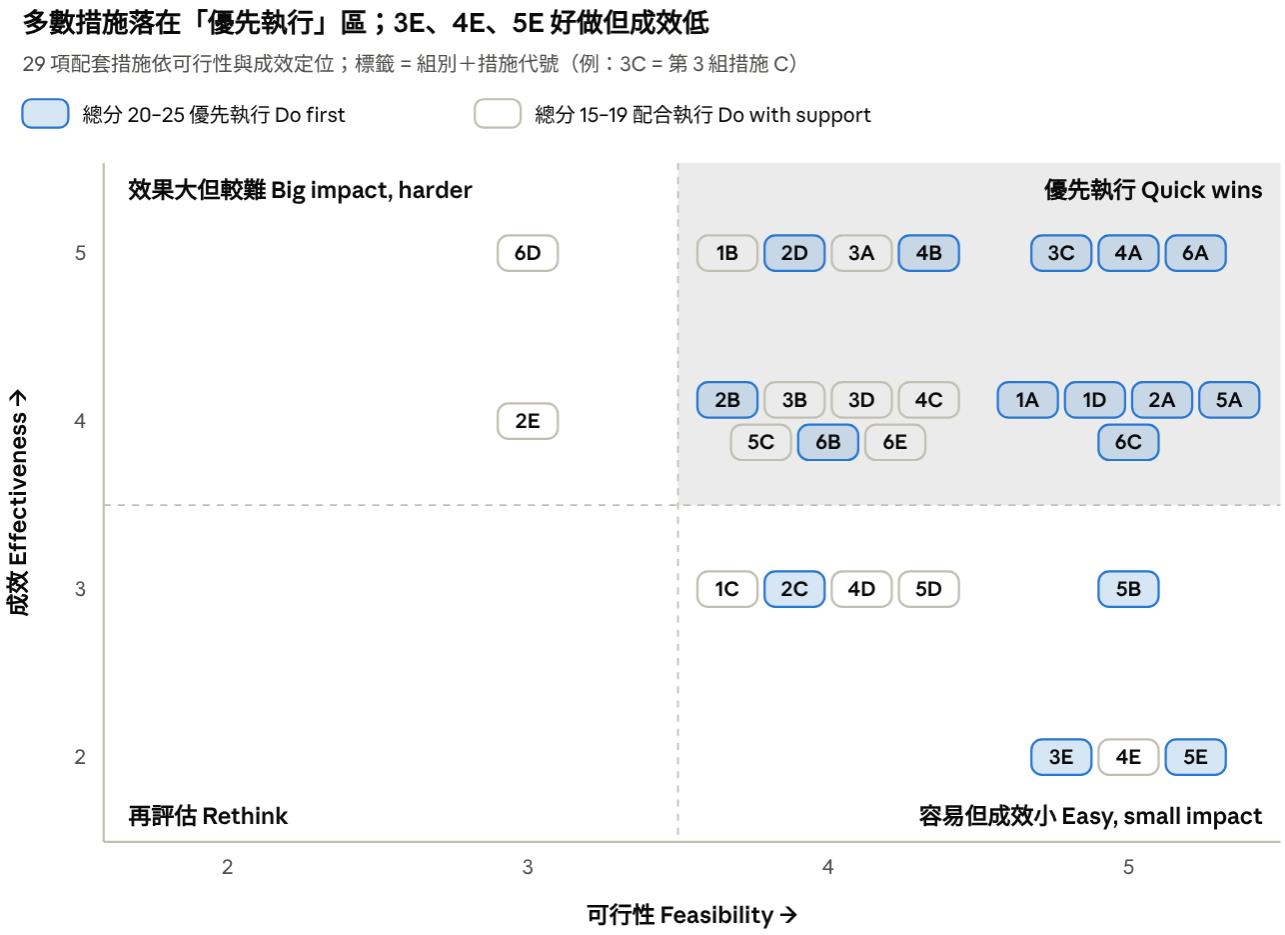
配套措施 Measure	可行性 F	成效 E	成本 C	時程 T	支持度 S	總分 /25	優先 Priority
C 賽後捐贈學校與社區 Donate to schools and communities	5	4	5	4	5	23	優先 Do first
A 設備租賃而非購買 Rent equipment instead of buying	5	5	4	4	4	22	優先 Do first
B 資產身分證（QR Code／RFID） Asset ID tags (QR code / RFID)	4	4	4	4	4	20	優先 Do first
E 循環條款寫入採購合約 Take-back rules in every contract	4	4	4	3	3	18	配合 Do with support
D 臨時場館模組化、可移地重組 Modular venues rebuilt elsewhere	3	5	2	3	4	17	配合 Do with support

具體作法 Concrete Actions

1. 採購前先問「能不能租？」，帳篷、發電機、座椅、電腦優先向廠商租用。 Ask "Can we rent it?" before buying. Rent tents, generators, seats and computers first.
2. 每件購入的資材貼 QR Code，系統記錄材質、數量與「賽後去處」。 Put a QR code on every item. The system records its material, number and planned "next home."
3. 賽前 1 年開放學校、社福團體登記需求，運動器材、家具在賽後 3 個月內送達。 One year early, let schools and charities sign up for what they need. Sports equipment and furniture arrive within three months after the Games.
4. 合約加入「廠商回收責任」：旗幟、地毯、包材由廠商回收再製成新產品。 Add a take-back rule to contracts: suppliers collect banners, carpets and packaging and turn them into new products.
5. 賽後公布「資材去處報告」，列出重複使用、捐贈、回收、捨棄的比例。 After the Games, publish a report showing how much was reused, donated, recycled or thrown away.

關鍵風險 Key risk：100% 循環很難完全達成，食物殘渣與受損資材仍會產生廢棄物。建議目標寫成「資材與設備 100% 有去處、掩埋或無法再利用的比例低於 5%」。 100% reuse is very hard, because food waste and broken items still exist. A clearer goal is "every item has a next home, and less than 5% goes to landfill."

六組一起看 Comparing All Six Groups



六組 29 項配套措施 · 可行性 × 成效

29 項措施中有 16 項總分 20 以上；各組最先做的是 1A／1D、2A、3C、4A、5A、6C。最難做的是 2E（AI 剩食預測）與 6D（模組化場館），成效最低的是 3E、4E、5E。 16 of the 29 measures score 20 or more. Each group should start with its top measure. 2E and 6D are the hardest to do, while 3E, 4E and 5E have the smallest impact.

關鍵字彙 Key Vocabulary

English	中文	例句 Example
feasibility	可行性	The plan has high feasibility because we already have the buses.
effectiveness	成效	We judge effectiveness by how much carbon is cut.
stakeholder	利害關係人	Residents are important stakeholders in the Games.
net-zero	淨零（碳排）	Net-zero means we remove as much carbon as we produce.

English	中文	例句 Example
carbon footprint	碳足跡	A beef meal has a bigger carbon footprint than a tofu meal.
carbon offset	碳抵換	Carbon offsets should be the last step, not the first.
greenwashing	漂綠	Saying "green" without real action is greenwashing.
renewable energy	再生能源	Solar and wind are renewable energy.
circular economy	循環經濟	In a circular economy, old things become new resources.
modular	模組化的	Modular homes can be moved and built again.
shuttle	接駁車	The shuttle takes athletes from the hotel to the stadium.
legacy	遺產；留下的成果	A good legacy helps the city long after the Games.